

CE 31500: TERM PROJECT

Computational Modeling of Groundwater Contamination Across New York State

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Abstract

In this report, groundwater contaminant transport was modeled for three representative soil conditions corresponding to Suffolk, Dutchess, and Cattaraugus Counties in New York. Numerical simulations were performed in Python using the FiPy library on a 150 m one-dimensional domain divided into 600 equal sections, with an initial Gaussian contaminant plume centered at $x = 5$ m, a source width of 2 m, zero-concentration boundary conditions at both ends of the domain, a total simulation time of 400 days, and a time step of 0.5 days. The results indicated that Suffolk County produced the greatest plume displacement, with a peak distance of 102.38 m and a peak concentration of 0.183440, Dutchess County showed an intermediate response with a peak distance of 11.38 m and a peak concentration of 0.615344, and Cattaraugus County showed the least plume displacement and the highest local concentration, with a peak distance of 5.62 m and a peak concentration of 0.933433. Overall, higher-permeability soils allowed contaminant plumes to travel farther and spread more widely, while lower-permeability soils kept contamination more localized but at higher concentrations near the source.

1 Introduction

Groundwater contamination has become an increasing issue in civil and environmental engineering due to industrialization and urban development (Freeze and Cherry, 1979, p. 8). Since pollutants can travel through soil, they can adversely affect drinking water supplies and pose risks to human health (U.S. EPA, 2015, p. C-8). For example, chemicals from leaking underground storage tanks can seep into the ground and contaminate nearby water sources. A conceptual representation of this process is shown in Figure 1.

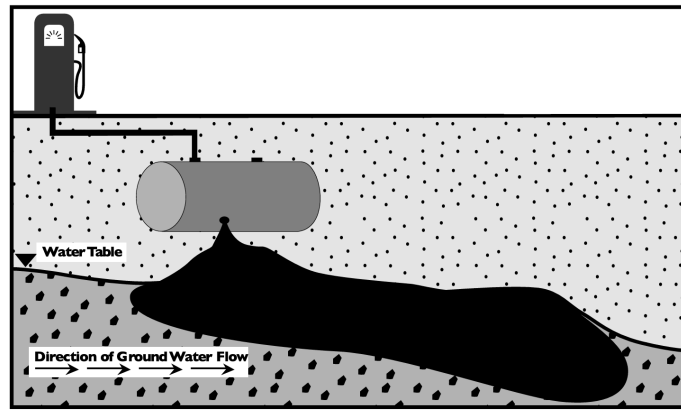


Figure 1: Groundwater contaminant plume from a leaking storage tank (U.S. EPA, 2015, p. C-1).

The movement of contaminants through soil can be described using the advection-dispersion equation (Freeze and Cherry, 1979, pp. 389–390), which accounts for both the flow of the fluid and the spread of contaminants within the soil. Advection refers to the movement of contaminants with the flowing groundwater, while dispersion describes the spreading of the contaminant plume caused by the differences in fluid velocity within the void spaces of the soil. These effects are influenced by properties such as particle size and permeability, meaning that soils with higher permeability generally allow faster contaminant movement and greater dispersion than lower permeability soils.

These transport mechanisms can be analyzed by numerically approximating the governing equation and solving it using computational methods. Finite volume methods can be used to solve the advection-dispersion equation by approximating contaminant transport over small control volumes in space and time. This provides a practical way to show how contaminants move through different soil types without requiring costly field experimentation. This approach builds upon the numerical methods introduced in class, but extends beyond the material by applying the derived model to representative soil conditions based on published soil information.

Groundwater contaminant movement depends heavily on the type of soil (Freeze and Cherry, 1979, pp. 10–11), meaning that different geographical regions will experience different transport properties under the same flow conditions and initial contaminant concentration. Coarse-grained soils, such as sand, typically allow groundwater to flow more rapidly and promote greater plume spreading, while fine-grained soils, such as clay and silt, tend to restrict flow and dispersion. In coarse-grained soils, contaminants can travel farther from the source and threaten a larger area of the groundwater system, requiring wider monitoring, containment, and remediation efforts. In fine-grained soils, contamination may remain concentrated near the injection site, increasing local pollutant levels and creating more severe impacts near the source. These differences highlight the importance of soil characterization for accurately predicting contaminant movement, assessing environmental impact, and selecting appropriate mitigation and groundwater protection strategies.

In this project, the advection-dispersion equation is solved numerically using the Python package FiPy, which applies the finite volume method to solve partial differential equations (Guyer et al., 2009, p. 6). The model is used to analyze groundwater contaminant movement in three different soil conditions for Suffolk, Dutchess, and Cattaraugus Counties in New York. Comparing these three cases provides insight into how groundwater contamination may spread in different environments and demonstrates how numerical methods can be applied to real-world engineering problems.

2 Methods

Numerical Model Setup

Groundwater contamination was modeled using the 1-D advection-dispersion equation, which describes the movement and spreading of contaminants within the soil. The governing equation is given by:

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} - v \frac{\partial C}{\partial x}, \quad (1)$$

where C is the contaminant concentration, t is time, x is position, D is the dispersion coefficient, and v is the groundwater velocity (Freeze and Cherry, 1979, p. 389). C was treated as a normalized, dimensionless concentration variable. The initial contaminant plume was represented using a Gaussian distribution centered at $x = 5$ m with a source width of $\sigma = 2$ m, and the concentration profile was then computed numerically over space and time by solving the advection-dispersion equation.

Numerical simulations were performed using the Python package FiPy, which is capable of solving partial differential equations using the finite volume method (Guyer et al., 2009, p. 6). The governing advection-dispersion equation was solved numerically over a one-dimensional computational domain representing subsurface flow (Freeze and Cherry, 1979, p. 352). The computational domain had a length of 150 m and was divided into 600 sections, giving a spatial step size of 0.25 m. The model was run for a total simulation time of 400 days with a time step of 0.5 days, resulting in 800 solution steps, and zero-concentration boundary conditions were applied at both ends of the domain. The Python code used to solve this equation is provided in Appendix C.

The FiPy package was selected because it provides a straightforward way to solve partial differential equations numerically and was ideal for implementing the one-dimensional advection-dispersion equation used in this project. A finite volume approach was chosen because it is related to the numerical methods introduced in class and is appropriate for tracking how contaminant concentra-

tion changes over space and time in a simple one-dimensional domain. Several simplifying assumptions were made in order to keep the study manageable. The model was limited to one-dimensional movement, groundwater velocity and dispersion were assumed to be constant within each soil case, the hydraulic gradient was fixed at $i = 0.01$, which was used in the groundwater velocity relationship $v = \frac{Ki}{n}$ described in Appendix B, and the contaminant concentration was treated as normalized rather than as an exact measured concentration.

These assumptions were justified because the goal of this project was to compare plume behavior across different soil conditions rather than to model a specific contaminated site in full detail. With these approximations, the model is expected to provide reasonable results to compare plume migration and spreading, but not exact predictions.

One programming difficulty was ensuring that the original and refined simulations could be compared consistently when estimating numerical error. This was addressed by storing the peak concentration from the original model for each soil case and then rerunning the model with a finer mesh and smaller time step so that the percent difference between the two solutions could be calculated directly.

Parameter Estimation and Assumptions

Three different soil conditions were used to represent different permeability and transport behavior. These conditions were selected to represent typical soil behavior associated with Suffolk, Dutchess, and Cattaraugus Counties in New York (USDA Soil Conservation Service, 1969, p. 42–46; USDA NRCS, 2001, pp. 168–169; USDA NRCS, 2007, p. 306). The assigned values are summarized in Table 1.

County	Soil Type	Velocity, v (m/day)	Dispersion, D (m ² /day)
Suffolk	Sandy / Rapid Permeability	0.244	0.122
Dutchess	Silty / Moderate Permeability	0.0158	0.00792
Cattaraugus	Clay / Slow Permeability	0.00144	0.000720

Table 1: Soil classifications and model values of groundwater velocity and dispersion used for Suffolk, Dutchess, and Cattaraugus County.

Representative values for groundwater velocity and the dispersion coefficient were estimated based on the relative permeability of each soil type (Bear, 1988, pp. 132–133; Freeze and Cherry, 1979, p. 26). Soils with higher permeability were assigned higher velocities and dispersion coefficients, while soils with lower permeability were assigned smaller values. The reference tables and calculations used to estimate these values are provided in Appendices A and B.

The resulting concentration profiles were analyzed by comparing the final plume shape, peak concentration, and distance traveled for each county soil condition at the end of the simulation. Particular attention was given to how far the plume migrated from the source location and how much the peak concentration decreased as the contaminant spread through the domain. A lower peak concentration with greater travel distance indicated stronger plume spreading, while a higher peak concentration near the source indicated more limited migration and less dispersion.

Error Estimation

Sources of error in the model were primarily associated with the numerical approximation used to solve the advection-dispersion equation and with the simplified parameter estimates assigned to each soil condition. The numerical error was estimated by performing a refinement check using a finer mesh of 1200 sections and a smaller time step of 0.25 days. The refined peak concentration was then compared to the original result for each county. The percent error was calculated using the equation

$$\text{Percent Error} = \frac{|\text{Refined Value} - \text{Original Value}|}{\text{Refined Value}} \times 100, \quad (2)$$

using the peak concentration from each simulation. The calculated errors are summarized in Table 3.

3 Results

The numerical simulations were used to generate contaminant concentration profiles after 400 days, as shown in Figure 2. The resulting curves were used to evaluate how the differences in soil permeability affect the movement and spreading of the contaminant plume.

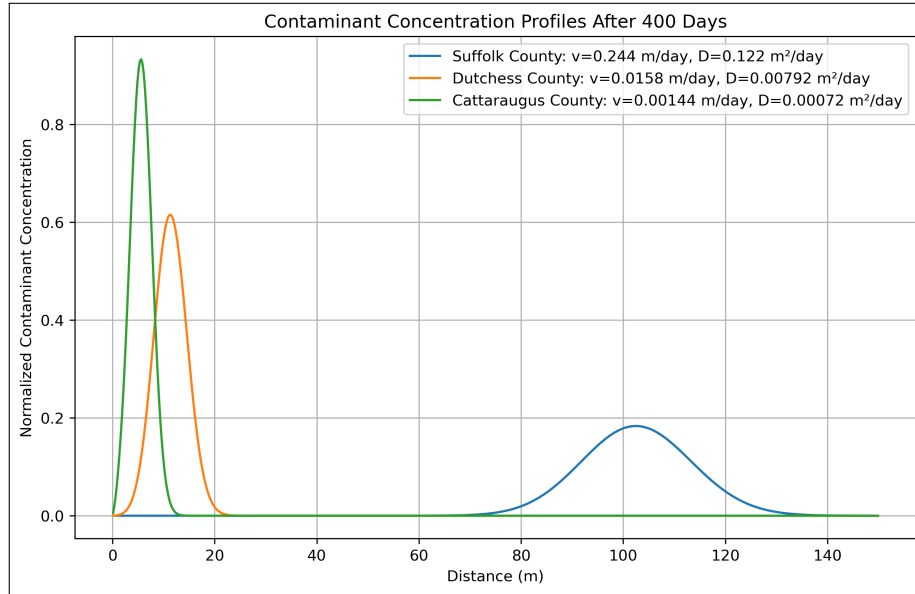


Figure 2: Contaminant concentration profiles after 400 days.

The peak concentration and corresponding peak distance for each county are summarized in Table 2.

County	Peak Concentration	Peak Distance (m)
Suffolk	0.183440	102.38
Dutchess	0.615344	11.38
Cattaraugus	0.933433	5.62

Table 2: Peak concentration and distance for each county after 400 days of simulated contaminant transport.

Contaminant Transport Behavior by County

From the results, Suffolk County experienced the greatest plume displacement. The concentration profile was located farthest from the source, with a peak distance of 102.38 m, showing that the contaminant traveled the greatest distance over the 400-day period. This behavior is expected given the higher groundwater velocity and dispersion coefficient used for the soil type determined for Suffolk County. In addition to moving farther, the plume also spread over a wider range and showed a lower peak concentration of 0.183440 compared to the other two counties. As a result, the contaminant is less concentrated at any single location. This reduces the severity of exposure at one point, but the wider plume increases the area of affected groundwater and places a larger population at risk.

Dutchess County showed an overall intermediate response. The concentration profile showed moderate spreading, with a peak distance of 11.38 m. This behavior allowed the contaminant to spread at a lower rate than in Suffolk while still producing noticeable plume movement away from the source. The peak concentration for Dutchess was 0.615344, which is greater than in the Suffolk case but lower than in the Cattaraugus case. Compared to Suffolk County, the contaminant remained more concentrated. This increases the severity of exposure at impacted locations, and its moderate plume spread still allows groundwater contamination to spread beyond the immediate source area. This response indicates a moderate level of risk, where both the contaminant concentration and affected area are significant, but it does not reach the wider plume extent observed in Suffolk County or the higher localized concentration observed in Cattaraugus County.

Cattaraugus County exhibited the least plume displacement. The concentration profile remained closest to the source, with a peak distance of 5.62 m. In addition to traveling the shortest distance, this model showed the narrowest plume spread and the highest peak concentration of 0.933433 among the three counties. As a result, the contaminant remained highly concentrated near the

source location. This increases the severity of exposure in the immediate area of contamination, but the limited plume displacement reduces the area of affected groundwater and decreases the overall number of people impacted compared to Suffolk and Dutchess soil conditions.

The overall results show a clear relationship between soil permeability and contaminant transport behavior. In highly permeable soils, the contamination affects a larger portion of the groundwater system, meaning that monitoring wells and other protective measures will need to cover a wider area. In lower-permeability soils, the contamination remains near the injection site, but the higher concentration may require more immediate and targeted remediation depending on the type of contaminant introduced to the soil. These results emphasize that soil conditions must be considered when predicting groundwater contamination and selecting strategies for groundwater protection, risk reduction, and environmental remediation.

Sources of Error and Model Limitations

Several sources of error may have influenced the numerical results obtained in this study. First, each county was simplified into one representative soil condition, even though the actual soil properties vary throughout each region. Since real soil conditions are more heterogeneous, the selected soil classifications may not fully capture the variations in permeability and porosity. Additionally, the groundwater velocity and dispersion coefficient used in the model were estimated from representative permeability classes and porosity values rather than direct site measurements. As a result, the selected values provide a simplified approximation of contaminant transport behavior rather than an exact prediction for each county.

Another limitation is that the model assumes one-dimensional transport under constant flow conditions. In real groundwater systems, contaminant movement may occur in multiple directions and may also be influenced by layered soils, changes in hydraulic gradient, chemical reactions, and other site-specific

factors. These effects were not included in the model. Therefore, the results should be interpreted as a comparison of generalized transport behavior across different soil conditions rather than a fully site-specific groundwater simulation.

Numerical error is also introduced through the way in which space and time are approximated in the model. The solution depends on the selected domain length, mesh size, and time step used in the FiPy model. Although these values were chosen to produce stable and reasonable concentration profiles, small changes in the predicted plume location and peak concentration may occur if smaller step sizes are used. Despite these limitations, the model still provides a useful representation of how soil permeability influences contaminant movement and spreading in groundwater systems.

The numerical error was estimated by performing a refinement check in which the model was rerun using a finer mesh and smaller time step. The percent difference between the original and refined peak concentrations for each county is summarized in Table 3.

County	Original Peak Concentration	Refined Peak Concentration	Percent Error (%)
Suffolk	0.183440	0.189825	3.36
Dutchess	0.615344	0.619296	0.64
Cattaraugus	0.933433	0.934605	0.13

Table 3: Error estimates from the refinement check comparing the original and refined peak concentrations for each case.

These results show that the numerical solution introduced only a small amount of numerical approximation error overall, with the Suffolk County case showing the greatest sensitivity, which can be attributed to its higher groundwater velocity and dispersion.

4 Conclusion

In conclusion, groundwater contaminant transport was modeled for three representative soil conditions corresponding to Suffolk, Dutchess, and Cattaraugus Counties in New York. Using the 1-D advection–dispersion equation and numerical simulations performed in Python with FiPy, contaminant con-

centration profiles were created to evaluate how soil permeability affects plume movement and spreading over time.

The results show that Suffolk County produced the greatest plume displacement, with a peak distance of 102.38 m and a peak concentration of 0.183440 after 400 days. In contrast, Cattaraugus County remained the most concentrated near the injection region, with the smallest peak distance of 5.62 m and the highest peak concentration of 0.933433. Dutchess County showed an intermediate response, with a peak distance of 11.38 m and a peak concentration of 0.615344. These differences show that soil permeability strongly affects both the distance a contaminant plume can travel and the extent to which it spreads throughout the groundwater system.

Overall, this highlights the importance of considering soil conditions when predicting groundwater contamination and when planning strategies for groundwater protection, risk reduction, and environmental remediation. Although the model used simplified soil conditions and estimated parameter values, it still provides a useful numerical comparison of contaminant movement across different regions.

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Appendices

Appendix A: Reference Tables

Table 2.5.1 Typical Porosity Values of Natural Sedimentary Materials^a			
Sedimentary Material	Porosity Value (percent)	Sedimentary Material	Porosity Value (percent)
Peat soil	60–80	Fine-to-medium mixed sand	30–35
Soils	50–60	Gravel	30–40
Clay	45–55	Gravel and sand	30–35
Silt	40–50	Sandstone	10–20
Medium-to-coarse mixed sand	35–40	Shale	1–10
Uniform sand	30–40	Limestone	1–10

^a After Todd 1959.

Figure A1: Values used to select porosity values for the model (Bear, 1988, p. 46).

Permeability. The quality of the soil that enables water or air to move downward through the profile. The rate at which a saturated soil transmits water is accepted as a measure of this quality. In soil physics, the rate is referred to as “saturated hydraulic conductivity,” which is defined in the “Soil Survey Manual.” In line with conventional usage in the engineering profession and with traditional usage in published soil surveys, this rate of flow continues to be expressed as “permeability.” Terms describing permeability, measured in inches per hour, are as follows:	
Impermeable	less than 0.0015 inch
Very slow	0.0015 to 0.06 inch
Slow	0.06 to 0.2 inch
Moderately slow	0.2 to 0.6 inch
Moderate	0.6 inch to 2.0 inches
Moderately rapid	2.0 to 6.0 inches
Rapid	6.0 to 20 inches
Very rapid	more than 20 inches

Figure A2: Permeability class definitions used to select hydraulic conductivity values for the model (USDA NRCS, 2007, p. 569).

Appendix B: Parameter Estimation

Representative values of groundwater velocity, v , and dispersion coefficient, D , were estimated using simplified soil classifications for Suffolk, Dutchess, and Cattaraugus Counties. The groundwater velocity was estimated from:

$$v = \frac{Ki}{n}, \quad (3)$$

where K is hydraulic conductivity, i is the hydraulic gradient, and n is porosity (Freeze and Cherry, 1979, p. 70). The hydraulic gradient was assumed to be $i = 0.01$ as a representative value so that all three counties could be compared under the same flow conditions. The dispersion coefficient was estimated using:

$$D \approx \alpha_L v, \quad (4)$$

where $\alpha_L = 0.5$ m was taken as the longitudinal dispersivity, and molecular diffusion was neglected (Freeze and Cherry, 1979, p. 390).

The representative soil conditions and selected values were taken from the midpoint values shown in Figures A1 and A2. These values (USDA NRCS, 2001, pp. 168–169; USDA NRCS, 2007, p. 306; USDA Soil Conservation Service, 1969, pp. 42–46) are listed below:

- Suffolk: sandy / rapidly permeable, $K = 13$ in/hr, $n = 0.325$
- Dutchess: silt loam / moderate permeability, $K = 1.3$ in/hr, $n = 0.50$
- Cattaraugus: silty clay / slow permeability, $K = 0.13$ in/hr, $n = 0.55$

Using the conversion

$$1 \text{ in/hr} = 0.6096 \text{ m/day},$$

the hydraulic conductivity values become:

$$K_{\text{Suffolk}} = 7.9248 \text{ m/day}$$

$$K_{\text{Dutchess}} = 0.79248 \text{ m/day}$$

$$K_{\text{Cattaraugus}} = 0.079248 \text{ m/day}.$$

Substituting into the velocity equation gives:

$$v_{\text{Suffolk}} = \frac{(7.9248)(0.01)}{0.325} = 0.244 \text{ m/day}$$

$$v_{\text{Dutchess}} = \frac{(0.79248)(0.01)}{0.50} = 0.0158 \text{ m/day}$$

$$v_{\text{Cattaraugus}} = \frac{(0.079248)(0.01)}{0.55} = 0.00144 \text{ m/day} .$$

Then,

$$D_{\text{Suffolk}} = (0.5)(0.244) = 0.122 \text{ m}^2/\text{day}$$

$$D_{\text{Dutchess}} = (0.5)(0.0158) = 0.00792 \text{ m}^2/\text{day}$$

$$D_{\text{Cattaraugus}} = (0.5)(0.00144) = 0.000720 \text{ m}^2/\text{day} .$$

Therefore, the final model values were:

County	v (m/day)	D (m ² /day)
Suffolk	0.244	0.122
Dutchess	0.0158	0.00792
Cattaraugus	0.00144	0.000720

Appendix C: Python Code for the Model

```

1  # Import necessary libraries
2  import numpy as np
3  import matplotlib.pyplot as plt
4  from fipy import CellVariable, Grid1D, TransientTerm,
   DiffusionTerm, ConvectionTerm
5
6  # Model values
7  soil_cases = {
8      "Suffolk County": {"v": 0.244, "D": 0.122},
9      "Dutchess County": {"v": 0.0158, "D": 0.00792},
10     "Cattaraugus County": {"v": 0.00144, "D": 0.000720}
11 }
12
13 # Domain setup
14 L = 150.0
15 nx = 600
16 dx = L / nx
17
18 mesh = Grid1D(dx=dx, nx=nx)
19 x = np.array(mesh.cellCenters[0])
20
21 # Time setup
22 total_time = 400
23 dt = 0.5
24 steps = int(total_time / dt)
25
26 # Initial plume setup
27 source_location = 5
28 source_width = 2
29
30 plt.figure(figsize=(10, 6))

```

```

31 | original_peaks = {}
32 |
33 |
34 | for name, values in soil_cases.items():
35 |     v = values["v"]
36 |     D = values["D"]
37 |
38 |     C = CellVariable(name="Concentration", mesh=mesh, value
39 |                       =0.0)
40 |     C.setValue(np.exp(-((x - source_location) ** 2) / (2 *
41 |                 source_width ** 2)))
42 |     C.constrain(0.0, where=mesh.facesLeft)
43 |     C.constrain(0.0, where=mesh.facesRight)
44 |     eq = TransientTerm() == DiffusionTerm(coeff=D) -
45 |         ConvectionTerm(coeff=(v,))
46 |     for step in range(steps):
47 |         eq.solve(var=C, dt=dt)
48 |
49 |     original_peaks[name] = np.max(np.array(C))
50 |
51 |     plt.plot(x, np.array(C), label=f"{name}: v={v} m/day, D
52 |           = {D} m^2/day")
53 |
54 | plt.xlabel("Distance (m)")
55 | plt.ylabel("Normalized Contaminant Concentration")
56 | plt.title("Contaminant Concentration Profiles After 400 Days
57 | ")
58 | plt.legend()
59 | plt.grid(True)
60 | plt.savefig("ModelGraph.png", dpi=300, bbox_inches="tight")
61 | plt.show()
62 |
63 | # Refined check for error estimate
64 | nx2 = 1200
65 | dx2 = L / nx2
66 | dt2 = 0.25
67 | steps2 = int(total_time / dt2)
68 |
69 | for name, values in soil_cases.items():
70 |     v = values["v"]
71 |     D = values["D"]
72 |
73 |     mesh2 = Grid1D(dx=dx2, nx=nx2)
74 |     x2 = np.array(mesh2.cellCenters[0])
75 |     C2 = CellVariable(name="Concentration", mesh=mesh2,
76 |                       value=0.0)
77 |     C2.setValue(np.exp(-((x2 - source_location) ** 2) / (2 *
78 |                 source_width ** 2)))
79 |     C2.constrain(0.0, where=mesh2.facesLeft)
80 |     C2.constrain(0.0, where=mesh2.facesRight)
81 |     eq2 = TransientTerm() == DiffusionTerm(coeff=D) -
82 |         ConvectionTerm(coeff=(v,))
83 |     for step in range(steps2):
84 |         eq2.solve(var=C2, dt=dt2)
85 |
86 |     peak_original = original_peaks[name]
87 |     peak_refined = np.max(np.array(C2))
88 |     percent_error = abs(peak_refined - peak_original) /
89 |         peak_refined * 100
90 |
91 |     print(name)
92 |     print("Original peak concentration =", peak_original)
93 |     print("Refined peak concentration =", peak_refined)
94 |     print("Percent error =", percent_error, "%")
95 |     print()

```

Listing 1: Python code used to solve the groundwater contaminant transport model.